

Streaming Service Customer Retention Analysis

Kanan Sharma

MKTG 768

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INTRODUCTION & OBJECTIVE

This report analyzes customer retention behavior for a major streaming service. The dataset includes information on customer streaming hours and whether the customer renewed or did not renew their subscription. The primary goal of this analysis is to understand which factors influence customer churn and when customers are most likely to stop renewing.

To study retention patterns, survival analysis techniques were used. The total number of hours streamed was treated as the time variable, and non-renewal was treated as the event. This approach allows us to analyze not only who churns, but also how churn risk changes as customer engagement increases.

Several variables were considered in the model, including customer acquisition method (trial offer and coupon use), engagement behavior (shows started, shows finished, days per week watching), and viewing characteristics (timeslot and pandemic registration status). These variables were selected because they logically influence subscription renewal decisions.

VARIABLES CONSIDERED IN THE MODEL

To understand customer loss, variables were selected based on three logical categories: acquisition method, engagement level, and customer viewing behavior.

Acquisition-related variables included whether the customer joined through a TV trial offer and whether a coupon code was used at signup. These factors help evaluate whether promotional strategies influence long-term retention.

Engagement variables included number of shows started, number of shows finished, and days per week watching. These measures capture how actively customers use the platform, which is expected to influence renewal behavior.

Customer context variables included sex, whether the customer registered during the pandemic period, and typical viewing timeslot. These factors help determine whether different behavioral segments display different churn risks.

These variables collectively allow us to evaluate both behavioral and promotional influences on customer retention.

OVERALL RETENTION PATTERN

The Kaplan–Meier survival curve shows the probability that a customer remains subscribed as streaming hours increase. The curve indicates that churn risk increases gradually as total streaming hours accumulate, with noticeable declines occurring after moderate usage levels.

This suggests that retention risk is not concentrated only in the very early period after signup. Instead, churn risk continues to evolve as customers accumulate streaming hours. This finding implies that retention efforts should not focus only on the first few days of signup but should also monitor customers as their engagement patterns develop.

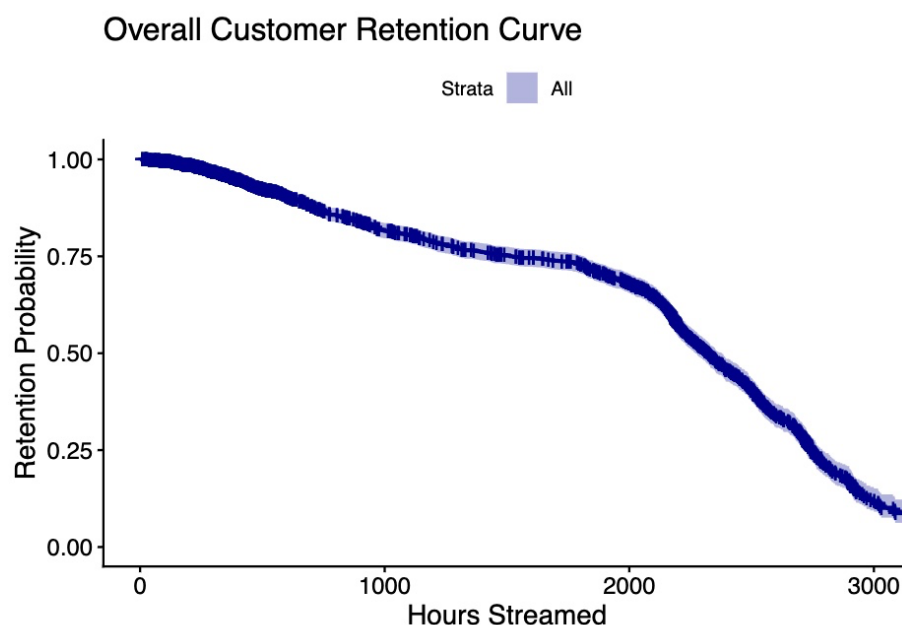


Figure 1. Overall probability of customer retention across streaming hours.

COX MODEL RESULTS: WHICH FACTORS IMPACT CHURN

To quantify the impact of each variable while controlling for others, a Cox proportional hazards model was estimated. The model showed statistically significant results overall (Likelihood Ratio test $p < 0.001$), indicating that the selected variables meaningfully explain variation in churn risk. The model's concordance index was 0.637, suggesting moderate predictive ability and performance above random classification.

Several variables had significant effects on non-renewal risk.

Customers who registered during the pandemic had a 38.9% higher risk of churn compared to those who registered outside that period. This suggests that pandemic-era users may have joined due to temporary circumstances and may be less committed long-term subscribers.

Each additional show started increased churn risk by approximately 5.5%. This may indicate that customers who frequently start new content but do not fully commit may be sampling rather than deeply engaging.

In contrast, each additional day per week of watching reduced churn risk by about 4%. This is a strong indicator that consistent weekly engagement promotes retention. Customers who integrate the platform into their weekly routine are more likely to renew.

Viewing timeslot also had a strong impact. Customers who primarily watched during late or off-peak time periods showed significantly higher churn risk. For example, certain timeslot groups had churn risk more than 70% to 100% higher than the reference category. This suggests behavioral segmentation plays an important role in predicting renewal.

Sex and coupon usage were not statistically significant predictors in the adjusted model, indicating that these variables alone do not strongly influence churn once engagement behavior is accounted for.

```
## Call:
## coxph(formula = Surv(hoursstreamed, event) ~ sexlismale + customerfromtvtrialoffer +
## couponcodeusedforinitialsignup + noofshowstarted + noofshowsfinished +
## pandemicregistration + daysperweekwatching + timeslotwatched,
## data = model_data)
##
## n= 1776, number of events= 888
##
##               coef exp(coef) se(coef)      z Pr(>|z|)
## sexlismale      0.02707   1.02744  0.06805   0.398  0.69077
## customerfromtvtrialoffer 0.16618   1.18079  0.08667   1.917  0.05519 .
## couponcodeusedforinitialsignup 0.06822   1.07060  0.09884   0.690  0.49005
## noofshowstarted  0.05402   1.05550  0.01373   3.933 8.37e-05 ***
## noofshowsfinished 0.08535   1.08910  0.07058   1.209  0.22656
## pandemicregistration 0.32861   1.38903  0.11455   2.869  0.00412 **
## daysperweekwatching -0.04171   0.95914  0.01714  -2.434  0.01495 *
## timeslotwatched2  0.33730   1.40116  0.25249   1.336  0.18158
## timeslotwatched3  0.53448   1.70655  0.23647   2.260  0.02380 *
## timeslotwatched4  0.70349   2.02078  0.29064   2.420  0.01550 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##               exp(coef) exp(-coef) lower .95 upper .95
## sexlismale      1.0274   0.9733    0.8992   1.1740
## customerfromtvtrialoffer 1.1808   0.8469    0.9963   1.3994
## couponcodeusedforinitialsignup 1.0706   0.9341    0.8821   1.2995
## noofshowstarted  1.0555   0.9474    1.0275   1.0843
## noofshowsfinished 1.0891   0.9182    0.9484   1.2507
## pandemicregistration 1.3890   0.7199    1.1097   1.7387
## daysperweekwatching 0.9591   1.0426    0.9275   0.9919
```

```

## timeslotwatched2      1.4012      0.7137      0.8542      2.2983
## timeslotwatched3      1.7066      0.5860      1.0736      2.7127
## timeslotwatched4      2.0208      0.4949      1.1432      3.5720
##
## Concordance= 0.637 (se = 0.01 )
## Likelihood ratio test= 109.5 on 10 df,  p=<2e-16
## Wald test              = 122.5 on 10 df,  p=<2e-16
## Score (logrank) test = 127.7 on 10 df,  p=<2e-16

```

Figure 2: Cox proportional hazards model showing hazard ratios, confidence intervals, and statistical significance.

DIFFERENCE BETWEEN USERS WHO STREAMED MORE THAN 1000 HOURS

A Kaplan-Meier comparison between customers who. Streamed more than 1000 hours and those who streamed fewer than 1000 hours revealed a highly significant difference ($p < 0.0001$).

Customers who exceeded 1000 streaming hours demonstrated substantially higher retention probability across the observed time horizon. The survival curve for heavy users remains consistently above that of lower usage customers.

This finding confirms that deep engagement strongly predicts renewal. Customers who accumulate high streaming hours are significantly less likely to churn. Encouraging customers to reach high-usage thresholds early in their lifecycle may be an effective retention strategy.

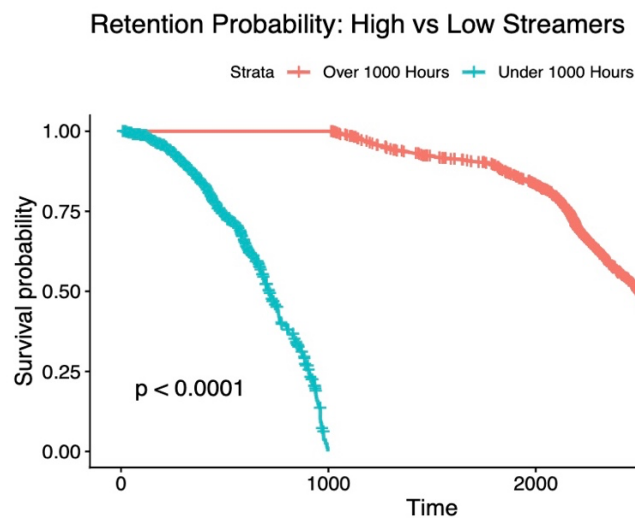


Figure 3: Retention comparison between customers streaming more than 1000 hours and those streaming less.

SHOULD THE COMPANY CONTINUE USING COUPON CODES AND TRIAL PERIODS?

Coupon usage did not significantly reduce churn risk in the adjusted model. The hazard ratio associated with coupon usage was close to 1 and statistically insignificant, suggesting that coupon-driven acquisition alone does not guarantee long-term retention.

Customers acquired through TV trial offers showed slightly higher churn risk, although the effect was only marginally significant. This suggests that while trial programs may increase initial signups, they may attract more price-sensitive or exploratory users who are less likely to renew.

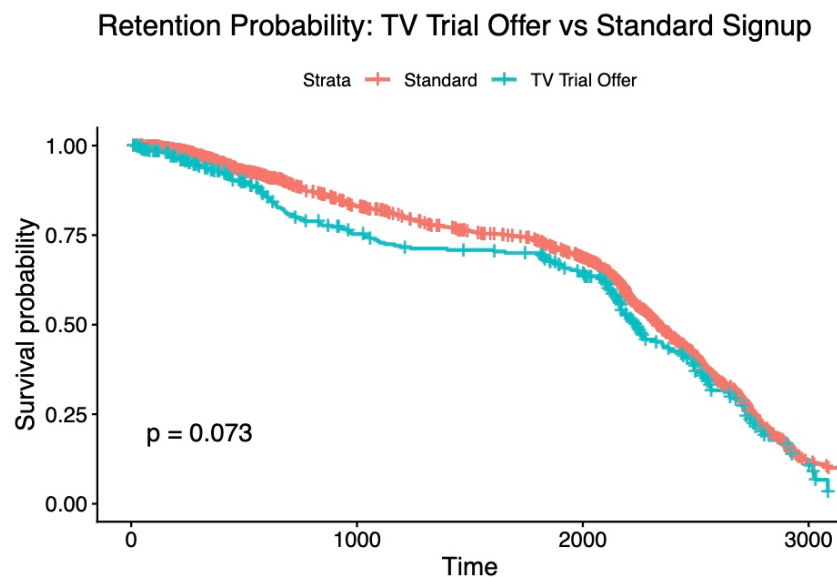


Figure 4: Retention comparison between TV trial offer users and standard signup customers. Log-rank $p = 0.073$.

Interpretation of Fig. 4: The survival curves show that customers who joined through a TV trial offer display slightly lower retention probabilities over time compared to standard signup customers. However, the difference is not statistically significant at the 5% level ($p = 0.073$). This suggests that while trial users may trend toward higher churn, the evidence is not strong enough to conclude a clear difference.

This does not mean coupons and trials should be eliminated. Instead, the findings suggest that promotional acquisition strategies should be paired with strong engagement-building efforts. Simply offering discounts is not sufficient; the platform must ensure that trial users develop consistent weekly usage habits before the trial ends.

A targeted follow-up strategy for trial users, such as personalized recommendations or engagement reminders, may improve long-term conversion rates.

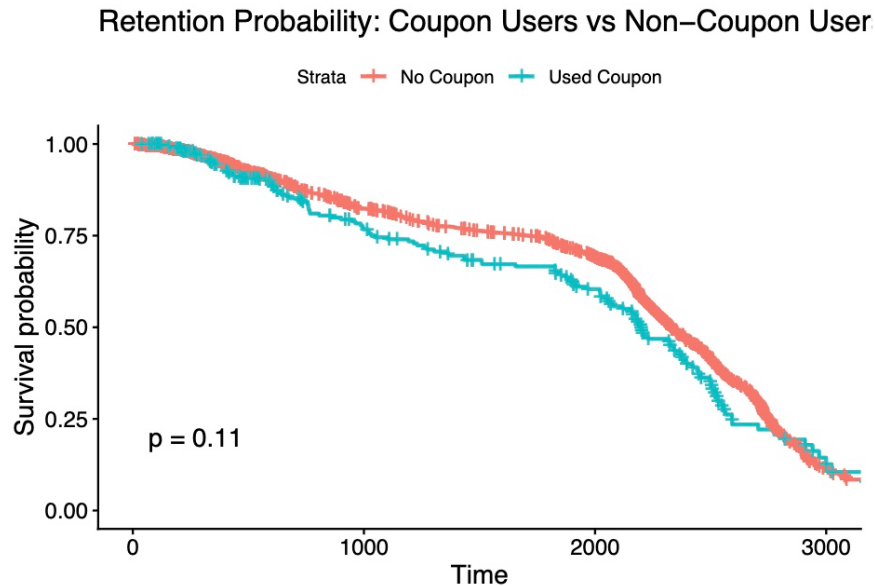


Figure 5 Retention comparison between coupon users and non-coupon users. Log-rank $p = 0.11$.

Note: The Kaplan–Meier curves are consistent with the Cox model findings, reinforcing that engagement variables play a stronger role in retention than promotional acquisition methods.

PRACTICAL RECOMMENDATIONS FOR THE STREAMING SERVICES

First, focus on increasing weekly engagement. Since days per week watching significantly reduces churn risk, the company should prioritize features that encourage consistent viewing behavior. Examples include personalized recommendations, watch streak reminders, or curated weekly content drops.

Second, monitor customers who registered during the pandemic or through trial offers more closely. These groups show higher churn risk and may benefit from proactive engagement outreach before renewal dates.

Third, encourage customers to complete shows rather than simply start new ones. The pattern observed suggests that surface-level exploration may not translate into long-term commitment. Content recommendation algorithms could emphasize series completion.

Fourth, create retention triggers around usage milestones. Since customers who exceed 1000 hours demonstrate significantly stronger retention, targeted messaging could encourage customers to reach engagement thresholds.

Finally, promotional strategies should shift from pure acquisition toward engagement-focused conversion. Trials and coupons can remain useful tools, but their success depends on how effectively the service converts trial behavior into habitual viewing.

LIMITATIONS

This analysis is based on observational data and therefore does not establish causation. Promotional offers may have been targeted toward specific customer segments, which could influence observed effects. The use of streaming hours as the time variable assumes that engagement accumulation is an appropriate measure of customer lifecycle progression. Additionally, while the model demonstrates moderate predictive performance (C-index = 0.637), there remains unexplained variation in churn behavior.

Future analyses could incorporate additional behavioral variables, subscription tenure measures, or content genre preferences to improve predictive accuracy.

CONCLUSION

Customer retention in this streaming service is strongly influenced by engagement behavior. Consistent weekly watching significantly reduces churn risk, while certain acquisition groups and viewing patterns are associated with higher non-renewal. Heavy users demonstrate substantially stronger retention, reinforcing the importance of encouraging deep platform usage.

The primary takeaway is clear: long-term retention is driven more by sustained engagement than by initial promotional incentives. The most effective strategy for reducing customer loss is to convert early usage into habitual, recurring viewing behavior.

This suggests that the streaming service should prioritize strategies that transform early engagement into consistent viewing habits rather than relying primarily on discount-based acquisition tactics.

